

“Constructing Representative Microdata with Sub-National Weights: A Two-Stage Approach Combining Machine Learning Imputation and Sparse Optimization”

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- **Policy needs:** Analysis of national legislation increasingly demands answers at sub-national geographies (States, Congressional Districts).
- **Data limitation:** National surveys are rarely designed to support direct small-area estimation (sample sizes thin down to a few hundred households). Lack of a single survey that contains all the variables required to the analysis and to run microsimulation models.
- **Dimensional paradox:** Microsimulation models require joint distributions rich enough to simulate e.g. complex tax rules, but files must remain small enough to run interactively.

Twofold Microdata Challenge

The FILL Problem

- No single survey observes all required policy variables together.
- Imagine you want to test a new tax law. Survey A (e.g. a general household survey) tells you how many children a family has and what their jobs are, but not its financial profile. Survey B (e.g. a wealth survey) tells you exactly how much money a family has in the bank, but not its family structure.
- Need to data fusion procedure to “glue” the missing information from survey B (“donor”) into survey A (“recipient”), creating complete family profiles.

The REPRESENT Problem

- Reduce the dataset size to a “manageable” dimension.
- Weight the enhanced file so it can reproduce granular administrative aggregates.
- Hierarchical & nested consistency constraints (National → State → District).

MICROCOSM (PolicyEngine) — An open-source microdata pipeline articulated in two stages.

- ➊ **Stage 1** offers a compelling case study in **modern data fusion**, repositioning the classical statistical matching problem within a **non-parametric machine learning** framework.
The empirical analysis fuses two primary data sources to create a comprehensive household dataset.
The process involves a recipient dataset (which has the household demographic data) and a donor dataset (which has rich financial details).
- ➋ **Stage 2** reduces the comprehensive dataset to a smaller certified release, ensuring that the remaining households match administrative aggregates (targets) across the country.

Stage 1: Design-Aware Imputation Model I

- The **Recipient dataset**: The US Census Bureau's Current Population Survey (CPS ASEC, public-use file), enhanced by PolicyEngine (eCPS) by pooling three ASEC vintages (2022–2024, ageing the two earlier vintages forward to the latest year): 337,704 records. It provides a highly representative sample of the US population with detailed information on demographics, labor supply, and cash incomes, but it lacks comprehensive data on wealth and household balance sheets.
- The **Donor dataset**: The Survey of Consumer Finances (SCF), administered by the Federal Reserve.
- The **Bridging variables** (Shared Features): The block of predictor variables common to both datasets. These include:
 - Demographics: Age, educational attainment, marital status, and household size.
 - Income streams: Wage income, self-employment income, capital gains, interest/dividend income, and social security benefits.

Stage 1: Design-Aware Imputation Model II

- The **Imputed variables** (The Targets): The goal of Stage 1 is to predict and inject the missing SCF financial variables into every household profile in the eCPS.
- The “glue”: **Quantile Regression Forest (QRF)**. The QRF acts as a predictive model that estimates the full distribution of potential target values, rather than a single average. By retaining the entire data distribution within each decision tree node, the model identifies specific percentiles—such as the 10th, 50th, or 90th—to simulate realistic population variations.

Why a ML approach? Traditional data fusion methods like nearest-neighbor hot-deck excel at maintaining the integrity of real-world data, including accurate outliers and zero values, because they copy existing records rather than creating synthetic data. However, these methods fail when using multiple, complex predictors (the curse of dimensionality), as finding exact matches across many variables becomes impossible. There is a trade-off between accurately preserving real data and managing complex, multi-dimensional datasets.

Stage 1: Design-Aware Imputation Model I

Three crucial improvements of MICROCOSM model to overcome standard Regression Forests problems:

- **Weighted Bootstrap:** Resamples training rows with replacement proportional to survey weights before growing trees. Corrects the “weight-blindness” flaw (Standard machine learning models ignore these weights).
- **The “Gate” for Zeros (Regime Gates):** Economic data is full of zeros. If you use a standard model to predict wealth, it tends to average things out. It might give everyone a tiny amount, completely wiping out the zeros. The model uses a regime gate. (In an econometric world: a selection model). The first step decides if the answer should be zero or not zero. If it decides “not zero,” a second forest calculates the exact amount.

Stage 1: Design-Aware Imputation Model II

- **Sequential Chaining:** Making the Balance Sheets Make Sense. If you have to guess multiple financial details for the same household (e.g., its debts, its assets, and its net worth), you cannot guess them completely separately. The model predicts the variables in a chain (one after the other). First, it guesses debt; then, conditioning on the debt, it guesses assets; finally, it calculates net worth based on the previous answers. This ensures accounting identities.

Stage 1: Comments I

- **Major strength:** It demonstrates that ignoring survey weights degrades net worth (Wasserstein) distance sixfold. Injected bootstrap successfully resolves this.
- **The non-extrapolation trap:** Quantile Regression Forests cannot extrapolate beyond training support boundaries. As a result, the model understates the q_{99} extreme tail by 20–30%. Possible to patch the QRF tree model with parametric Pareto tail distributions to capture the missing 20–30% top-percentile wealth mass?
- **Comparisons:** Empirically comparing the stage 1 imputation model with traditional hot-deck imputation would clearly justify the use of a more complex machine learning approach.

- **Synthetic data fusion as the inevitable starting point?** We should consider the alternative paradigm of exact or probabilistic Record Linkage. Why not physically link the survey data to real administrative tax records at the individual level? The roadblock here has always been legal privacy and data confidentiality constraints. Yet, recent breakthroughs in Privacy-Preserving Record Linkage (PPRL) —using cryptographic techniques— now allow researchers to link distinct databases without ever exposing personal identifiable information.

Stage 2: Target-Informed Sparse Selection

- Stage 1 has created complete profiles for the 337,704 households. **The dataset is way too big and heavy for microsimulations.**
- Stage 2 of MICROCOSM reduces the data support while forcing calibration alignment.
- Alignment to 32,633 granular administrative benchmarks (targets) for tax, benefits, and demographics.
- **The optimization process uses hard concrete gates to retain 57,240 records that accurately match these government targets**, ensuring the final dataset aligns with real-world administrative records.

Stage 2: The administrative targets

The **32,633 administrative targets** represent the total number of fiscal, socioeconomic, and demographic aggregates (compiled from official United States institutional databases) used to calibrate the microsimulation model.

- These 32,633 targets are macro-level data points nested across three geographic layers in the United States. This “surface” consists of:
 - 24,340 congressional-district targets (individual electoral districts).
 - 7,815 state-level targets.
 - 478 national-level targets.
- These aggregates come from authoritative administrative sources. The vast majority—31,350 targets—come from the IRS Statistics of Income (SOI). The remaining targets are drawn from the Census Bureau (population estimates) and registries of public assistance programs (such as Medicaid, SNAP/food stamps, and TANF).
- They mix count constraints (e.g., the number of program recipients in a specific district) and dollar amount constraints (e.g., total tax revenue or expenditures).

Stage 2: Target-Informed Sparse Selection

The targets serve three core purposes:

- ➊ **Ensuring sub-national accuracy** (Spatial Reweighting): The model shifts and fine-tunes household weights so that the microdata aggregates scale up to match known, local administrative reality simultaneously at the national, state, and district levels.
- ➋ **Driving smart data pruning** (Target-Informed Selection): Instead of dropping records at random, the pipeline uses a sparse optimization method. The algorithm actively evaluates the records against the 32,633 targets: a specific household record survives the pruning process only if keeping it directly helps the model "hit" a target that would otherwise be missed.
- ➌ **Enabling highly detailed policy simulation**: Because the final dataset is tightly calibrated against this massive surface of real-world benchmarks, MICROCOSM can accurately simulate the impact of new tax and benefit legislation (such as the **One Big Beautiful Bill Act** analyzed in a companion paper) down to the specific grain of individual states and congressional districts.

Stage 2: The Optimization Objective (Loss Function)

The optimizer minimizes a multi-term objective function (L) to balance matching the targets, keeping the file small, and avoiding erratic weights:

$$L = L_{\text{cal}} + \lambda_{L0} L_0 + \lambda_{L2} L_2$$

- L_{cal} (**Calibration Loss**): This measures how close the simulated totals are to the real administrative targets. It uses a metric called capped weighted MAPE (Mean Absolute Percentage Error). It calculates relative error (so a 1% miss on a small target costs the same as a 1% miss on a large target) but caps extreme outliers so single hard-to-fit targets do not corrupt the gradients.
- L_0 (**Sparsity Penalty**): This term penalizes the number of active records to force the dataset to be small.
- L_2 (**Concentration Penalty**): This prevents the optimizer from "cheating" by placing massive, unrealistic weights on just a few households to hit the targets.

Stage 2: Solving the L_0 Problem: Stochastic Gating

- The algorithm is given a double mission. It must minimize the error against real-world administrative data, but it also gets "punished" for every household switch (on/off) it leaves open.
- To force the algorithm to actually use those switches and delete records, the authors add a penalty called the L_0 norm, which counts non-zero entries in a vector.
- However, the L_0 norm is non-differentiable, meaning you cannot use standard gradient descent to solve it because you cannot calculate its derivatives.
- To bypass this, the authors implement Hard Concrete gates ($z_i \in [0, 1]$) attached to each household parameter: Instead of a hard "yes/no" choice, the gate is treated as a random variable drawn from a Concrete distribution (a continuous relaxation of a discrete choice).
- This "trick" allows to turn continuous optimization into discrete record selection.

Stage 2: The Post-Selection Refit

- The weights generated while the switches are turning on and off are not very accurate. The file gets a high error score of 9.86%.
- The authors separate the process into two steps. First, they use the gates purely as a filter to choose the best support subset of households (57,240 record). Second, lock that selection in, throw away the stochastic gates, and run a standard dense Adam optimizer for 1,500 epochs to smoothly refit ordinary calibration weights onto those selected records.
- This drops the final error to an impressive 4.74%.

Empirical Results: Full-Surface Comparison

Table: Full-surface comparison after 1,500-epoch fits (32,633 administrative targets)

Method Arm	Records	Objective Loss ↓	Median ARE	ESS ↑	Max Weight
Informed L_0 + Refit	57,240	4.74%	0.89%	4,726	913,836
Dense no- L_0 Calibration	337,704	5.07%	0.56%	5,970	579,298
Random Sampling + Reweight	57,240	7.55%	6.70%	2,480	1,163,939
Informed L_0 Gated Weights	57,240	9.86%	11.66%	6,879	546,455
Dense Random Scaled Sample	57,240	24.24%	55.24%	981	2,176,121

Core Success:

- L_0 + Refit lowers the loss by 6.4% over Dense and 37.2% over Random.

Visible Risks:

- Post-selection refit is non-negotiable.
- High maximum weight spikes.

The Evaluation Breakthrough: Population-View Harness

Traditional benchmarking checks input marginals. This paper scores candidate files by projecting them through multiple partial survey views against held-out samples:

- **Weighted Energy Distance:** Sample form of a strictly proper scoring rule to check full distributions.
- **PRDC Coverage:** Anti-collapse axis based on k -NN radius; invariant to candidate reweighting.
- **Classifier Two-Sample Test (C2ST):** Cross-validated AUC of a gradient-boosted judge trying to distinguish real vs. imputed rows.
- **Uncapped Tail Block:** Evaluates extreme heavy right tails (q_{90} and q_{99} ratios) without caps.

Stage 2: Comments I

- **The gated weight paradox.** The authors find that the continuous weights generated directly during the joint optimization loop yield a poor objective loss of nearly 10%. To fix this, they discard those weights and run an independent, secondary post-selection refit exclusively on the chosen support. If the continuous relaxed gates are poor weight-setters on their own, how confident can we be that they are truly selecting the optimal structural support? Does this behavior imply that a lasso alternative or a discrete combinatorial search might identify an even stronger subset?
- **Prove the L_0 superiority.** Section 6.6 of the paper critiques L_1 (LASSO) soft-thresholding shrinkage, but no L_1 empirical arm is actually run in the benchmarks. Run a proximal gradient L_1 comparison to numerically justify using the complex Hard Concrete distribution.

Stage 2: Comments II

- **Single-Seed vulnerability:** High-stakes numbers on the full target surface are derived from a single fixed random seed configuration. Is it possible to replace single-seed results with multi-seed variance blocks to demonstrate that the support selection is statistically stable?
- **Weight inflation** (Representative Fragility). Because the optimizer forces a small number of records (57,240) to match tens of thousands of targets, individual household weights skyrocket up to 913,836. Having one household represent nearly a million families, does it make the simulator volatile to minor policy tweaks?
- **Is compressing the dataset down to 57,240 records actually necessary for modern microsimulation?** The authors argue that a smaller file is required for computational efficiency and to run interactive microsimulations. In an era of rapid development in processing power, wouldn't it be far more methodologically sound to simulate policy rules directly on a full, unreduced dataset of 337,000 households, or even larger data sets,

thereby completely avoiding the statistical trade-offs and fragility caused by forced sparsity?